

VTT AMBR Cultivation Data

Comparison of 4-, 5-, and 6-variable representations

EXECUTIVE MESSAGE

All three representations recover structure in the 29 cultivation profiles. Adding Volume substantially improves cluster stability, while adding Off-gas CO₂% gives the strongest selected cluster separation and introduces additional, non-redundant information. However, the current notebook contains a selection-logic issue: it selects the **second-highest silhouette solution for all three representations**, not only for the 6-variable case. Therefore the reported $k = 5, 7$, and 3 solutions should be described as selected exploratory solutions rather than mathematical optima until the code is corrected and rerun.

Purpose. This report explains what changes when the same VTT AMBR cultivations are represented using four core variables, then Volume, then Off-gas CO₂%. It is written for a biological audience and focuses on what the statistics mean for interpretation of cultivation behaviour.

1. Dataset and analysis strategy

The comparison uses the same **29 reactor/run cultivation profiles** in all three representations. This is important: differences between results are therefore driven by the variables included, rather than by a changing set of experiments.

Representation	Variables	Initial features	After correlation filter
4 variables	Glucose, DO, pH, Biomass	24	17
5 variables	+ Volume	30	18
6 variables	+ Off-gas CO ₂ %	36	22

For each process variable, six trajectory descriptors were extracted: normalized area under the curve, maximum slope, maximum, minimum, initial value, and final value. Features with $|\text{correlation}| > 0.85$ were filtered to reduce redundancy. Remaining features were standardized, summarized by PCA, and clustered with K-means. Enough principal components were retained to explain at least 75% of total variance.

KEY FINDING - WHAT IT MEANS

The comparison is fair because all three representations contain the same 29 cultivations and use the same preprocessing. Adding a variable is therefore being tested as an information gain, not as a change in sample composition.

2. Dimensionality and information retained by PCA

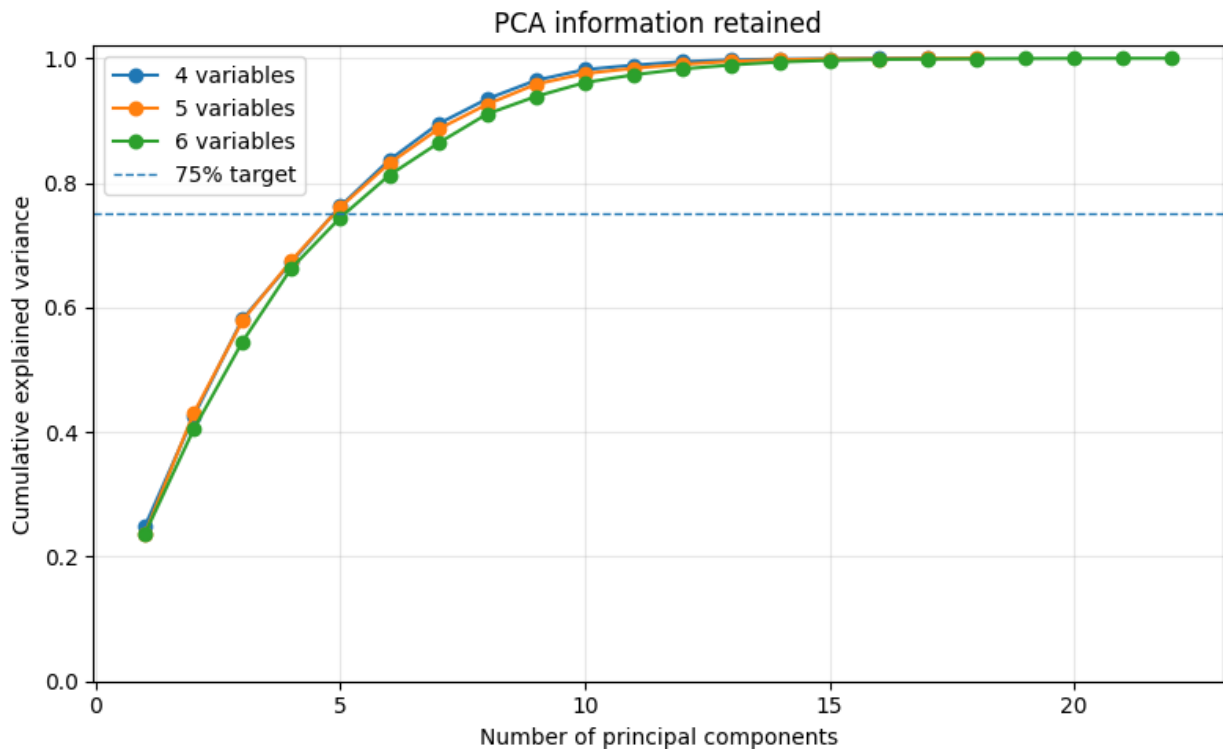


Figure 1. Cumulative variance explained by PCA for the three representations. The dashed line marks the 75% target used to define the clustering space.

The 4- and 5-variable representations each require **five principal components** to retain at least 75% of the information. The 6-variable representation requires **six components**. PC1 and PC2 explain 42.5%, 43.0%, and 40.4% of total variance for the 4-, 5-, and 6-variable representations, respectively.

The slightly lower two-dimensional variance for six variables is not evidence that the six-variable model is worse. It indicates that the additional information is distributed over more independent directions. In biological terms, Off-gas CO₂% may capture aspects of cultivation behaviour that are not fully aligned with glucose, oxygen, pH, biomass, or volume.

KEY FINDING - WHAT IT MEANS

Adding Off-gas CO₂% increases the dimensionality needed to summarize the cultures (6 PCs instead of 5). This is consistent with CO₂ carrying additional information rather than simply duplicating an existing measurement.

3. Cluster separation across variable sets



Figure 2. Silhouette score across candidate numbers of clusters. Higher values indicate groups that are more compact internally and better separated from one another.

Across all three representations, the silhouette curves show that **k = 2 gives the strongest raw separation** (approximately 0.42-0.43 from the plotted results). The notebook nevertheless reports selected exploratory solutions of k = 5 for four variables, k = 7 for five variables, and k = 3 for six variables, with silhouette scores of 0.376, 0.398, and 0.416, respectively.

Within these selected finer-grained solutions, separation improves progressively from four to six variables. The increase is modest, but the six-variable solution gives the highest selected silhouette score.

KEY FINDING - WHAT IT MEANS

At a broad level, the cultures naturally separate most strongly into two groups. If the scientific goal is to identify finer cultivation phenotypes, the selected multi-cluster solutions can still be useful, but they should be presented as exploratory subdivisions rather than the silhouette optimum.

4. Stability of the selected clustering solutions

Representation	Selected k	Silhouette	Subsampling ARI	ARI SD
4 variables	5	0.376	0.539	0.251
5 variables	7	0.398	0.862	0.137
6 variables	3	0.416	0.684	0.346

Cluster stability was assessed by repeatedly removing 20% of cultivations, reclustering the remaining experiments, and comparing the assignments with the full-data solution using the Adjusted Rand Index (ARI). An ARI close to 1 means that the grouping is reproducible after modest changes to the dataset.

The **5-variable representation is clearly the most stable** for its selected $k = 7$ solution (mean ARI 0.862, SD 0.137). The 6-variable $k = 3$ solution is moderately stable (ARI 0.684) but more variable between subsamples (SD 0.346). The 4-variable $k = 5$ solution is the least stable (ARI 0.539).

KEY FINDING - WHAT IT MEANS

Volume appears to add useful structure: the 5-variable representation produces the most reproducible fine-grained grouping. Adding CO2 improves separation, but the selected 3-cluster solution is less stable than the 5-variable solution and should therefore be interpreted with more caution.

5. What the PCA maps show biologically

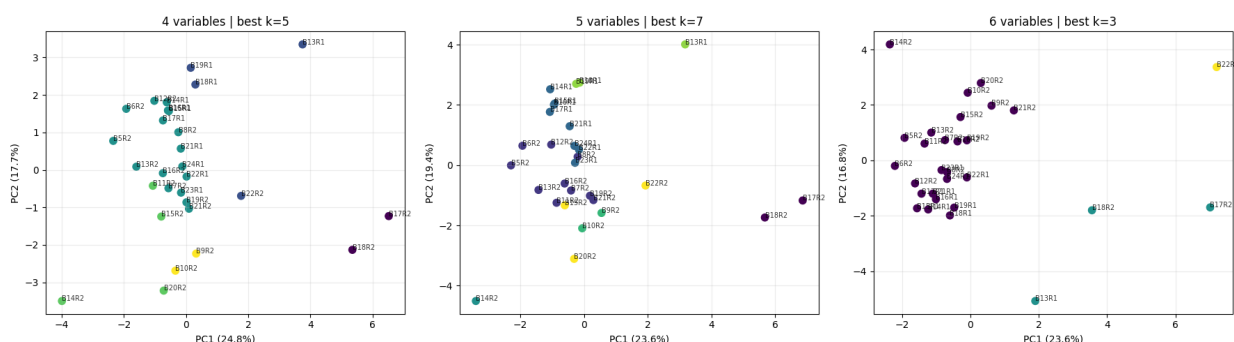


Figure 3. PC1-PC2 projections for the selected 4-, 5-, and 6-variable cluster solutions. Labels identify individual bioreactor/run cultivations.

The PCA maps show a central group of cultivations together with several profiles that consistently occupy more isolated regions. **B17R2 and B18R2** remain particularly distinct in the representations, while B14R2, B13R1 and B22R2 also become prominent in some variable sets.

In the six-variable representation, most cultivations belong to one broad group, while B17R2 and B18R2 form a distinct pair and B22R2 forms a separate small group. This suggests that the additional variables, especially Off-gas CO₂%, may accentuate specific process behaviours in a limited subset of cultivations rather than uniformly reorganizing all experiments.

KEY FINDING - WHAT IT MEANS

The main biological question is not only which cluster number scores best, but why specific cultures repeatedly separate from the main population. B17R2, B18R2, B22R2, B14R2 and B13R1 are priority candidates for comparison with experimental conditions and raw trajectories.

6. Do Volume and Off-gas CO₂% add new information?

Volume contributes meaningfully to the PCA representation. Its retained AUC feature has an absolute loading up to **0.403** among the first five PCs (largest contribution on PC2). This indicates that volume variation is not completely redundant with the original four variables.

Several Off-gas CO₂% descriptors survive correlation filtering and contribute across the first six PCs. The strongest absolute loadings are approximately **0.372 for final CO₂**, 0.309 for CO₂ AUC, 0.288 for maximum CO₂ slope, and 0.287 for minimum CO₂. Because these descriptors load on several PCs, CO₂ appears to contribute distributed information rather than a single duplicated axis.

KEY FINDING - WHAT IT MEANS

Both added variables contribute information after redundancy filtering. Volume is associated with a strong and stable reorganization of the feature space, while Off-gas CO₂% adds a complementary metabolic signal that improves selected-cluster separation but also increases dimensionality.

7. Agreement between the three clustering solutions

Comparison	Adjusted Rand Index
4 vs 5 variables	0.468

Comparison	Adjusted Rand Index
4 vs 6 variables	0.289
5 vs 6 variables	0.126

The selected cluster assignments are not equivalent across variable sets. The 4- and 5-variable solutions show moderate agreement (ARI 0.468), whereas agreement is lower when the six-variable solution is compared with either of the others. Part of this is expected because the selected numbers of clusters differ (5, 7, and 3). It nevertheless shows that adding variables changes how similarity between cultivations is represented.

KEY FINDING - WHAT IT MEANS

The choice of variables materially changes the biological grouping. Clusters should therefore not yet be treated as fixed biological classes. They are hypotheses about similar cultivation behaviour that need validation against metadata and raw time-series profiles.

8. Overall interpretation and choice of representation

Criterion	4 variables	5 variables	6 variables
Parsimony	Strong	Moderate	Lowest
Selected silhouette	0.376	0.398	0.416
Selected stability (ARI)	0.539	0.862	0.684
PCs for >=75% variance	5	5	6
Added process information	Baseline	Volume	Volume + CO2

There is no single representation that dominates every criterion. The **5-variable representation** gives the most reproducible selected clustering, whereas the **6-variable representation** gives the highest silhouette among the selected fine-grained solutions and retains biologically interpretable CO2 information. The 4-variable representation is simpler but gives the weakest stability for the selected $k = 5$ solution.

For a biological report, the most defensible interpretation is therefore that **Volume provides clear added value**, while **Off-gas CO2% provides promising complementary metabolic information** whose effect should be validated against experimental metadata. If the purpose is robust subgroup identification, the 5-variable representation currently has the strongest stability evidence. If the purpose is richer mechanistic/process characterization, the 6-variable representation is attractive, provided the CO2-driven differences can be biologically explained.

KEY FINDING - WHAT IT MEANS

The results support keeping at least the 5-variable representation. The 6-variable representation is scientifically valuable because CO2 contributes independent metabolic information, but the final choice between 5 and 6 variables should be made after checking whether the CO2-associated groups correspond to known cultivation conditions or reproducible biological phenotypes.

9. Important technical note before finalizing cluster numbers

The current notebook code ranks silhouette values and then assigns **selected_k = ranked_k[1]** without conditioning this choice on the representation. As a result, the second-best silhouette solution is selected for **all three** variable sets. This explains why the summary reports $k = 5, 7$, and 3 even though Figure 2 shows that $k = 2$ has a higher silhouette score in each case.

If the intended rule is to use the silhouette optimum for the 4- and 5-variable analyses and the second-best solution only for the 6-variable analysis, the notebook should be corrected and rerun before the cluster numbers are presented as final. The feature counts, PCA variance, loading analysis, and silhouette curves remain informative; the affected outputs are the selected cluster assignments, stability estimates tied to those assignments, and cross-solution ARIs.

KEY FINDING - WHAT IT MEANS

This is a code-selection issue, not a failure of the dataset. Correcting it is necessary for a final statistical comparison. Until then, this report treats $k = 5, 7$, and 3 as deliberate exploratory partitions, not as the mathematically optimal cluster counts.

10. Recommended biological follow-up

The next analysis should connect the multivariate patterns back to biology. For the cultivations that repeatedly separate from the main population, compare the original glucose, DO, pH, biomass, volume, and off-gas CO₂% trajectories and then inspect cultivation metadata (medium, feeding strategy, aeration/agitation conditions, inoculum, run, and any known experimental deviations). This will establish whether the clusters correspond to meaningful process phenotypes rather than purely statistical structure.

KEY FINDING - WHAT IT MEANS

The statistical analysis identifies where the unusual cultivations are; the metadata and raw trajectories are needed to explain why. That biological validation is the step that can turn the clustering from an exploratory visualization into a useful process interpretation.

Data source: results and figures extracted from the executed notebook *VTT_Compare_4_5_6_Variables(1).ipynb*. No external biological assumptions were used to assign meaning to the clusters.